Timetables

1 Thilini's school timetable is shown below:

Start	Finish	Monday	Tuesday	Wednesday	Thursday	Friday
8:45 am	9:55 am	Maths	English	History	English	Geography
9:55 am	10:05 am	RECESS	RECESS	RECESS	RECESS	RECESS
10:05 am	11:20 am	Science	History	Science	Geography	P.E.
11:20 am	12:40 pm	History	Maths	Maths	P.E.	English
12:40 pm	1:20 pm	LUNCH	LUNCH	LUNCH	LUNCH	LUNCH
1:20 pm	2:30 pm	P.E.	Geography	Sport	English	Maths
2:30 pm	3:15 pm	Geography	Science	Sport	Maths	Science

a What lesson would Thilini have at 9:00 on Monday?

b How long was the Science lesson on Tuesday?

c How much time is allocated to Sport?

d How long is the school day at this school?

e How many more minutes are allocated to Maths than History?

f Thilini arrives late on Thursday due to a doctor's appointment. If he arrives halfway through lunch what time did he arrive?

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2 Consider the city train timetable below:

Station	pm	pm	pm	pm	pm	pm	pm
Museum	1:11	1:15	1:17	1:21		1:26	1:30
St. James	1:13		1:19	1:23	1:26		1:32
Circular Quay	1:17	1:20	1:23	1:27			1:36
Wynyard	1:19	1:22		1:29	1:31		1:38
Town Hall	1:22	1:25		1:32			1:41
Central	1:26	1:29	1:30	1:36	1:36	1:37	1:45

a If Charlie catches a train at 1:23 pm from Circular Quay, what time will he arrive at Central?

b How long does the 1:17 pm train from Museum take to get to Central?

c Adam arrives at Circular Quay at 1:29 pm and he needs to catch a train to go to Central. What time does the next train depart Circular Quay to go to Central?

d The 1:15 pm train from Museum arrived at Wynyard 6 minutes late. What time did it arrive?

e James is currently at work near Museum station. What is the latest time he can arrive at Museum station if he has to be at Town Hall station by 1:38 pm?

f If Susie is at Museum at 1:20 pm, what is the earliest time she can get to Central by?

3 Consider the bus timetable below:

Bus Stop	am	am	am	am	am	am
Pine St	4:33	5:36		7:30	8:32	9:31
Westbrook Dr		5:38		7:39	8:38	
Merry Oaks	4:40	5:41	6:36	7:42	8:47	9:43
Carlton Rd	4:47	5:42	6:43		8:50	9:40
Highlands Dr	4:51	5:48	6:51	7:48	8:51	9:48
West St		6:17	7:12	8:18	9:13	10:18

Kate travels to work by bus. Her home is a 10-minute walk from the bus stop at Merry Oaks, and her workplace is a 5-minute walk from the stop at Highlands Drive. She has to be at work by 8 am.

a What is the latest time Kate can leave home to get to work on time?

b If she misses her bus, how late will Kate be to work?

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4 Consider the light rail timetable below:

Central to Dulwich Hill

Monday to Friday					
Central, Sydney	08:53	every	13:53	-	14:05
Capitol Square, Haymarket	08:56	8 - 12 mins	13:56	-	14:08
Paddy's Market, Haymarket	08:58		13:58	-	14:10
Exhibition Centre, Sydney	09:01		14:01	-	14:13
Convention, Sydney	09:03		14:03	14:11	14:15
Pyrmont Bay, Pyrmont	09:04		14:04	14:12	14:16
The Star, Pyrmont	09:05		14:05	14:13	14:17
John Street Square, Pyrmont	09:06		14:06	14:14	14:18
Fish Market, Pyrmont	09:08		14:08	14:17	14:20
Wentworth Park, Pyrmont	09:09		14:09	14:18	14:21
Glebe, Glebe	09:10		14:10	14:19	14:22
Jubilee Park, Glebe	09:12		14:12	14:21	14:24
Rozelle Bay, Annandale	09:14		14:14	14:23	14:26
Lilyfield, Lilyfield	09:16		14:16	14:25	14:28
Leichhardt North, Lilyfield	09:18		14:18	14:27	14:30
Hawthorne, Leichhardt	09:20		14:20	14:29	14:32
Marion, Leichhardt	09:22		14:22	14:30	14:34
Taverners Hill, Leichhardt	09:23		14:23	14:32	14:35
Lewisham West, Lewisham	09:24		14:24	14:33	14:36
Waratah Mills, Dulwich Hill	09:26		14:26	14:35	14:38
Arlington, Dulwich Hill	09:27		14:27	14:35	14:39
Dulwich Grove, Dulwich Hill	09:28		14:28	14:36	14:40
Dulwich Hill, Dulwich Hill	09:29		14:29	14:38	14:41

a If you arrive at Wentworth Park at 13:21, what is the longest time that you are expected to wait for the next train?

b How long does the train typically take to get from Wentworth Park to Taverners Hill?

c Without any delays, what is the longest expected time it should take to get the light rail from Wentworth Park to Taverners Hill?

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5 Consider the train timetable below:

Penshurst to Bondi junction

Monday to Friday							
Penshurst	09:19	-	09:32	-	-	-	09:48
Hurstville	09:22	09:20	09:35	09:30	09:42	09:47	09:51
Allawah	-	09:23	-	09:33	-	-	09:53
Carlton	-	09:25	-	09:35	-	-	09:55
Kogarah	-	09:27	-	09:37	-	09:51	09:57
Rockdale	-	09:29	-	09:39	-	09:53	09:59
Banksia	-	09:31	-	09:41	-	-	10:01
Arncliffe	-	09:33	-	09:43	-	-	10:03
Wolli Creek	09:29	09:36	09:42	09:46	09:49	09:57	10:06
Tempe	-	09:37	-	09:47	-	-	10:07
Sydenham	09:33	09:40	09:46	09:50	-	10:00	10:10
Redfern	09:39	09:46	09:52	09:56	09:59	10:06	10:16
Central	09:42	09:49	09:55	09:59	10:02	10:09	10:19
Town Hall	09:45	09:52	09:58	10:02	-	10:12	10:22
Martin Place	09:47	09:54	10:00	10:04	-	10:14	10:24
Kings Cross	09:49	09:56	10:02	10:06	-	10:16	10:26
Edgecliff	09:52	09:59	10:05	10:09	-	10:19	10:29
Bondi Junction	09:55	10:02	10:08	10:13	-	10:22	10:32

a How many minutes does it take to get from Tempe to Martin Place if someone took the 09:47 train to Bondi Junction?

b If Michael arrives at Hurtsville station at 09:28, how long will he have to wait for the next train that stops at Wolli Creek?

c Which train should Michael catch to arrive at Wolli Creek the earliest? Explain your answer.

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6 Consider the bus timetable below:

City Circular Quay to Parramatta

Monday to Friday					
Circular Quay	14:38	14:50	-	15:00	15:10
Martin Place Station	14:44	14:56	-	15:06	15:16
Town Hall Station	14:50	15:02	-	15:12	15:22
Town Hall House	14:53	15:05	-	15:15	15:25
Victoria Rd at Hornsey St	14:58	15:10	-	15:20	15:30
Victoria Rd before Darling St	15:00	15:12	-	15:22	15:32
Victoria Rd after Lyons Rd	15:06	15:18	-	15:28	15:38
Riverside Girls High School	-	-	15:27	-	-
Trim Place	15:14	15:26	15:31	15:36	15:46
Blaxland Rd before Hatton St	15:24	15:36	15:41	15:46	15:56
Blaxland Rd opp Community Health Centre	15:25	15:37	15:42	15:47	15:57
Victoria Rd at Gaza Rd	15:35	15:47	15:52	15:57	16:08
Victoria Rd at Wharf Rd	15:41	15:53	15:58	16:03	16:14
Betty Cuthburt Ave Shops	15:45	15:57	16:02	16:07	16:18
Rydalmere Park	15:49	16:01	16:07	16:12	16:23
Victoria Rd before Railway St	15:53	16:05	16:11	16:16	16:27
Parramatta Station	16:04	16:16	16:22	16:27	16:38

a Jenny's friend drops her off at Trim Place station at 03:30 pm after school. She is going to her house which is a 6-minute walk from Parramatta station. What is the least amount of time she can be home in?

b Uther is getting on the bus at Victoria Rd at Gaza Rd to ride the bus with Jenny. What time should he get the bus so he will be on the same bus as Jenny? Give your answer in 24-hour time.

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7 Consider the ferry timetable below:

Watsons Bay to Pyrmont Bay

Monday to Friday							
Watsons Bay		14:49	15:19	15:49	16:19	-	-
Rose Bay		14:59	15:29	15:59	16:29	16:49	17:09
Circular Quay	ARR	15:12	15:42	16:12	16:42	17:02	17:22
Circular Quay	DEP	15:17	15:47	16:17	16:47	17:07	17:27
Milsons Point		15:22	15:52	16:22	16:52	17:12	17:32
McMahons Point		15:25	15:55	16:25	16:55	17:15	17:35
Balmain East		15:30	16:00	16:30	17:00	17:20	17:40
Barangaroo	ARR	15:36	16:06	16:36	17:06	17:26	17:46
Barangaroo	DEP	15:40	16:10	16:40	17:10	17:30	17:50
Pyrmont Bay		15:44	16:14	16:44	17:14	17:34	17:54

Saturday							
Watsons Bay		-	14:49	-	15:19	-	15:49
Rose Bay		-	14:59	-	15:29	-	15:59
Circular Quay	ARR	-	15:12	-	15:42	-	16:12
Circular Quay	DEP	15:02	15:17	15:32	15:47	16:02	16:17
Milsons Point		15:08	15:22	15:38	15:52	16:08	16:22
McMahons Point		-	15:25	-	15:55	-	16:25
Balmain East		-	15:30	-	16:00	-	16:30
Barangaroo	ARR	15:18	15:36	15:48	16:06	16:18	16:36
Barangaroo	DEP	15:22	15:40	15:52	16:10	16:22	16:40
Pyrmont Bay		15:26	15:44	15:56	16:14	16:26	16:44

a Which wharves will have most departing ferries on Saturdays at the times shown?

b If you arrive at Circular Quay on a weekday between 3 and 4 pm, what is the longest time you will have to wait to catch a ferry?

c If you arrive at Circular Quay on a Saturday between 3 and 4 pm, what is the longest time you will have to wait to catch a ferry?

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8 Paul wants to go on a 2-hour tour of the Vatican City and a 1-hour walking tour of the Roman Forum. He is not yet sure in which order he wants to attend the tours, but he needs to have at least 90 minutes to travel from one tour to the other. The table below shows the start times of each tour:

Vatican City Tour	9:00 am	10:00 am	12:30 pm	1:30 pm	2:30 pm	4:00 pm	5:30 pm
Roman Forum Tour	12:00 pm	1:00 pm	2:00 pm	3:00 pm	4:00 pm	5:00 pm	6:00 pm

a What is the earliest time Paul can start the Roman Forum tour if he does the Vatican City tour first?

b What time is the latest Vatican City tour Paul can take if he wants to finish the Roman Forum tour at 6 pm?

9 Caitlin is travelling across Sydney. She wants to catch the ferry from Manly Wharf to Circular Quay, then a train to Campbelltown. Caitlin wants to get to Campbelltown train station by 8:00 pm. It takes her 4 minutes to walk between the Circular Quay ferry wharf and the Circular Quay train station.

Journeys from Manly Wharf to Circular Quay

Ferry wharf						
Manly	5:18 pm	5:48 pm	6:18 pm	6:48 pm	7:18 pm	7:48 pm
Circular Quay	5:48 pm	6:18 pm	6:48 pm	7:18 pm	7:48 pm	8:18 pm

Journeys from Circular Quay to Campbelltown

Bus Station						
Circular Quay	5:55 pm	6:25 pm	6:55 pm	7:25 pm	7:55 pm	8:25 pm
Campbelltown	7:17 pm	7:47 pm	8:17 pm	8:47 pm	9:17 pm	9:47 pm

a What time is the latest train that Caitlin can catch to get to Campbelltown by 8:00 pm?

b What is the latest ferry that Caitlin can catch from Manly Wharf so that she arrives in Campbelltown before 8:00 pm?

c How long will it take Caitlin to travel between Manly Wharf and Campbelltown Station in total?

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10 Fiona is attending a lecture by a famous scientist at 9 pm on Thursday at Swansea University. She has to take a train from Gowerton to Abertawe, and then a bus from Abertawe bus station to the university. To get a good seat at the lecture, Fiona wants to arrive at least 10 minutes before the start time.

Journeys from Gowerton to Abertawe

Train Station						
Gowerton	4:17 pm	5:17 pm	6:17 pm	7:17 pm	8:17 pm	9:17 pm
Abertawe	4:31 pm	5:31 pm	6:31 pm	7:31 pm	8:31 pm	9:31 pm

Journeys from Abertawe to Swansea University

Bus Station						
Abertawe	6:36 pm	7:36 pm	8:36 pm	9:36 pm	10:36 pm	11:36 pm
University pm	6:50 pm	7:50 pm	8:50 pm	9:50 pm	10:50 pm	11:50 pm

a What time is the latest bus that Fiona can catch to get to Swansea University on time?

b Abertawe bus station is a 3-minute walk from Abertawe train station. What is the latest train Fiona can take from Gowerton to make sure she catches the right bus?

c How long will it take Fiona to travel between Gowerton and Swansea Universty in total?

11 Kathleen and Joanne want to go to the cinema to see The Matrix. They have arranged to meet at 9 am at the cinema on Albert Road.

Journeys toward Albert Road

Bus Stop						
Riley St	6:31 am	7:28 am	8:27 am	9:26 am	10:30 am	11:29 am
Rutland St	6:39 am	7:39 am	8:36 am	9:36 am	10:39 am	11:38 am
Lacey Rd	6:46 am	7:47 am	8:47 am	9:44 am	10:45 am	11:47 am
Albert Rd	6:57 am	7:56 am	8:57 am	9:55 am	10:56 am	11:57 am

Journeys from Albert Road

Bus Stop						
Albert Rd	4:31 pm	5:27 pm	6:27 pm	7:26 pm	8:31 pm	9:27 pm
Lacey Rd	4:39 pm	5:38 pm	6:36 pm	7:36 pm	8:40 pm	9:36 pm
Rutland St	4:46 pm	5:46 pm	6:47 pm	6:44 pm	8:46 pm	9:45 pm
Riley St	4:57 pm	5:55 pm	6:57 pm	7:55 pm	8:57 pm	9:55 pm

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a Kathleen lives 13 minutes from the bus stop at Riley Street. What is the latest time she can leave her house to get to Albert Road on time?

b Joanne lives 10 minutes from the bus stop at Lacey Road. What is the latest time she can leave her house to get to Albert Road on time?

c Kathleen must be home by 7:45 pm. If The Matrix runs for 90 minutes, what is the latest screening that they can watch?

Film	Session times (pm)
A Beautiful Mind	4:15, 6:30, 9:15
Good Will Hunting	5:15, 6:15, 7:15
Hidden Figures	4:30, 6:30, 9:15
The Matrix	4:15, 6:30, 8:15
Snakes on a Number Plane	5:15, 6:30, 8:30

12 Ivan and Sharon have planned a date together. They want to go to see Hidden Figures at the Central Cinema and have dinner at La Trattoria. The restaurant is 12-minute's walk from the cinema, and they want to give themselves at least 15 minutes to buy tickets and popcorn before the movie starts. La Trattoria says they can book a table for one hour either at 7:30 pm or 9 pm.

Film screening times

Film			
Good Will Hunting	8:45 pm	9:30 pm	10:15 pm
Hidden Figures	7:15 pm	9:30 pm	11:15 pm
The Matrix	8:30 pm	9:30 pm	10:45 pm
A Beautiful Mind	7:00 pm	8:15 pm	9:45 pm
Median Girls	7:00 pm	9:30 pm	11:30 pm

Train journeys to Central Station

Station				
New Street	6:35 pm	7:36 pm	8:32 pm	9:31 pm
Town Hall	6:43 pm	7:44 pm	8:40 pm	9:39 pm
Downtown	6:48 pm	7:49 pm	8:45 pm	9:44 pm
Central	6:55 pm	7:56 pm	8:52 pm	9:51 pm

a If they decide to have dinner first, what is the earliest time they can watch Hidden Figures?

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b Hidden Figures runs for 90 minutes. If they decide to watch the movie first, what is the latest screening time they can watch?

c Sharon and Ivan decide to have dinner first. They travel from New Street Station to Central Station by train. If the station is a 4-minute walk from La Trattoria, what is the latest train they can catch to get to their dinner reservation on time, and still have time to watch a screening of Hidden Figures?

13 Bill and Tom plan to meet at Manchester Piccadilly Station and then travel together to the Manchester United Football Ground Railway Station on Saturday. The train timetable is shown below:

Station					
Manchester Piccadilly	5:22 pm	5:45 pm	6:20 pm	6:44 pm	7:23 pm
Manchester Oxford Rd	5:30 pm	5:53 pm	6:28 pm	6:52 pm	7:31 pm
Manchester Utd FG	5:34 pm	5:57 pm	6:32 pm	6:56 pm	7:35 pm

a If the match kicks off at 7 pm, what is the latest train they can catch from Manchester Piccadilly Station to get to the match on time?

b Consider the following tram timetable of the eastern tram line:

Tram Stop					
Ashton	6:18 pm	6:23 pm	6:30 pm	6:45 pm	7:14 pm
Manchester Piccadilly	6:29 pm	6:34 pm	6:41 pm	6:56 pm	7:25 pm

Bill will travel from Ashton to Manchester Piccadilly Station by tram. What is the latest time he can leave Ashton to get to Manchester Piccadilly in time to catch the train from part (a)?

c Consider this tram timetable of the northern tram line:

Tram Stop					
Prestwich	6:05 pm	6:24 pm	6:46 pm	7:14 pm	7:35 pm
Market St	6:16 pm	6:35 pm	6:58 pm	7:25 pm	7:43 pm

Tom will travel from Prestwich Station to Market Street and then walk for 5 minutes to Manchester Piccadilly station. What is the latest tram he can catch from Prestwich Station to make the train from part (a)?

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14 Elizabeth is on holiday in Hong Kong. She wants to go by train from Kowloon Station to Disneyland Resort Station.

Morning journeys from Kowloon to Disneyland

Station						
Kowloon	7:17 am	7:29 am	8:19 am	8:31 am	9:16 am	9:28 am
Olympic	7:21 am	7:33 am	8:23 am	8:35 am	9:20 am	9:32 am
Nam Cheong	7:23 am	7:35 am	8:25 am	8:37 am	9:22 am	9:34 am
Lai King	7:27 am	7:39 am	8:29 am	8:41 am	9:26 am	9:38 am
Tsing Yi	7:31 am	7:43 am	8:33 am	8:45 am	9:30 am	9:42 am
Sunny Bay	7:38 am	7:50 am	8:40 am	8:52 am	9:37 am	9:49 am
Disneyland	7:45 am	7:57 am	8:47 am	8:59 am	9:44 am	9:56 am

Evening journeys from Disneyland to Kowloon

Station						
Disneyland	4:18 pm	4:30 pm	5:10 pm	5:22 pm	6:15 pm	6:27 pm
Sunny Bay	4:25 pm	4:37 pm	5:15 pm	5:29 pm	6:22 pm	6:34 pm
Tsing Yi	4:32 pm	4:44 pm	5:24 pm	5:36 pm	6:29 pm	6:41 pm
Lai King	4:36 pm	4:48 pm	5:28 pm	5:40 pm	6:33 pm	6:45 pm
Nam Cheong	4:40 pm	4:52 pm	5:32 pm	5:44 pm	6:37 pm	6:49 pm
Olympic	4:42 pm	4:54 pm	5:34 pm	5:46 pm	6:39 pm	6:51 pm
Kowloon	4:46 pm	4:58 pm	5:38 pm	5:50 pm	6:43 pm	6:55 pm

a How long will her train journey be?

b Elizabeth wants to spend at least 9 hours at Disneyland, but she must be back at Kowloon Station by 6 pm. What is the latest train Elizabeth can take from Kowloon to Disneyland Resort Station in the morning?

c If she went from Kowloon to Disneyland Resort by taxi instead, the journey would only take 15 minutes. How much time would Elizabeth save by taking a taxi to Disneyland Resort instead?

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15 Consider the tide chart below:

	Tuesday 25/10	Wednesday 26/10	Thursday 27/10	Fri 28/10
Low tide (morning)	4:34 am	5:31 am	6:25 am	7:26 am
High tide (morning)	10:47 am	11:42 am	12:12 am	1:11 am
Low tide (afternoon)	5:02 pm	6:05 pm	6:53 pm	7:52 pm
High tide (afternoon)	11:20 pm	-	12:40 pm	1:39 pm

a How long is there between the first high tide on Thursday and the next high tide?

b How long is there between the last low tide on Thursday and the next low tide?

c A boat can only enter the harbour within two hours before or after a high tide. If a fishing boat travels out of the harbour at 2:12 am on Thursday, when is the earliest it can return back to the harbour?

d What is the minimum length of time that the boat from part c could be out fishing for?

16 Consider the following chart that shows the rising and setting times for the Sun:

	Sunrise	Sunset	Daylight	Difference	Time	Altitude
24 May	7:17 am	6:02 pm	10 h 45 m 30 s	−48 s	12:40 pm	44.0°
25 May	7:17 am	6:02 pm	10 h 44 m 42 s	−47 s	12:40 pm	43.8°
26 May	7:18 am	6:02 pm	10 h 43 m 56 s	−46 s	12:40 pm	43.6°
27 May	7:18 am	6:02 pm	10 h 43 m 12 s	−44 s	12:40 pm	43.4°
28 May	7:19 am	6:01 pm	10 h 42 m 29 s	−43 s	12:40 pm	43.3°
29 May	7:19 am	6:01 pm	10 h 41 m 47 s	−41 s	12:40 pm	43.1°
30 May	7:20 am	6:01 pm	10 h 41 m 07 s	−40 s	12:40 pm	43.0°

a For how long between the 27th and 28th of May will there be no sunlight?

b Are the days getting longer or shorter?

c Aaron wants to watch the sunset from the end of a local pier. To see the full sunset he wants to arrive 30 minutes before the sunset and he is located at a 22 minute walk from the spot. What is the latest he can leave to see the sunset on May 28? Express your answer in 24-hour time.

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d Harvey leaves his house to go cycling on a bush track at 4:30 pm on the 26th of May. His headlight is broken and he knows the track becomes too dark for him to see easily at 20 minutes before sunset. If he heads in one direction along the path and returns the same way and at the same speed, what is the latest time he can turn around to come home? Express your answer in 12-hour time.

17 The following chart shows sunrise and sunset times from a coastal town in New South Wales, Australia:

Date	January			February			March	
	Sunrise	Sunset		Sunrise	Sunset		Sunrise	Sunset
1	0448	1907		0516	1859		0542	1831
2	0448	1907		0517	1858		0543	1830
3	0449	1907		0518	1857		0544	1828
4	0450	1907		0519	1857		0544	1827
5	0451	1908		0520	1856		0545	1826
6	0452	1908		0521	1855		0546	1825
7	0452	1908		0522	1854		0547	1823
8	0453	1908		0523	1853		0548	1822
9	0454	1908		0524	1852		0548	1821
10	0455	1908		0525	1851		0549	1820
11	0456	1907		0526	1851		0550	1818
12	0457	1907		0527	1850		0551	1817
13	0458	1907		0528	1849		0551	1816
14	0459	1907		0529	1848		0552	1814
15	0500	1907		0529	1847		0553	1813
16	0500	1907		0530	1846		0554	1812
17	0501	1906		0531	1845		0555	1810
18	0502	1906		0532	1844		0555	1809
19	0503	1906		0533	1842		0556	1808
20	0504	1905		0534	1841		0557	1806
21	0505	1905		0535	1840		0558	1805

(Source: Astronomical Applications Department of the U.S. Naval Observatory)

a What time is sunset on January 16? Express your answer in 12-hour time.

b For how long on March 2 is there sunlight?

c For how long between January 4 and 5 is there no sunlight?

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d If Laura wants to watch the sunrise from her favourite lookout, she needs to leave the house at least 16 minutes before the sunrise. What is the latest she can leave to see the sunrise on February 19? Express your answer in 12-hour time.

e Lachlan needs the sun to have set at least 2 hours prior to using his telescope. What is the earliest time that he can use his telescope on March 10? Express your answer in 12-hour time.

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